Maharashtra Education Society's
Institute of Management and Career Courses (IMCC), Pune
AUTONOMOUS

Academic Year 2026-27
FY MCA- SEM I
IT12 Data Structure and Algorithms
Practice Assignment1

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1. **Create Linked List**
2. **Treverse and print the node values**
3. **Insert node at a specific position**
4. **Find Middle node and print its value**
5. **Delete node**
6. **Reverse list**
7. **Calculate the sum of every two consecutive node values.**

#1. Create a Linked List in Python

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

n1 = Node(10)
n2 = Node(20)
n3 = Node(30)

n1.next = n2
n2.next = n3

temp = n1
while temp:
    print(temp.data, end=" -> ")
    temp = temp.next

print("None")
```

```
10 -> 20 -> 30 -> None
```

2. Traverse and Print the Node Values in Python

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

n1 = Node(10)
n2 = Node(20)
n3 = Node(30)

n1.next = n2
n2.next = n3

temp = n1
while temp:
    print(temp.data)
    temp = temp.next
```

```
10
20
30
```

3. Insert node at a specific position

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

head = Node(10)
head.next = Node(20)
head.next.next = Node(30)

data = int(input("Enter value: "))
pos = int(input("Enter position: "))

new = Node(data)

if pos == 1:
```

```
                new.next = head
    head = new
else:
    temp = head
    for i in range(pos - 2):
        temp = temp.next
    new.next = temp.next
    temp.next = new

temp = head
while temp:
    print(temp.data, end=" -> ")
    temp = temp.next
print("None")
```

```
Enter value: 15
Enter position: 2
10 -> 15 -> 20 -> 30 -> None
```

4. Find the Middle Node and Print Its Value in Python

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

head = Node(10)
head.next = Node(20)
head.next.next = Node(30)
head.next.next.next = Node(40)
head.next.next.next.next = Node(50)

slow = fast = head

while fast and fast.next:
    slow = slow.next
    fast = fast.next.next

print("Middle node:", slow.data)
```

```
Middle node: 30
```

```
# 5. Delete a Node from a Linked List in Python
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

head = Node(10)
head.next = Node(20)
head.next.next = Node(30)

pos = int(input("Enter position to delete: "))

if pos == 1:
    head = head.next
else:
    temp = head
    for i in range(pos - 2):
        temp = temp.next
    temp.next = temp.next.next

temp = head
while temp:
    print(temp.data, end=" -> ")
    temp = temp.next
print("None")
```

```
Enter position to delete: 2
10 -> 30 -> None
```

```
# 6. Reverse a Linked List in Python
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

head = Node(10)
head.next = Node(20)
head.next.next = Node(30)

prev = None
```

```
temp = head

while temp:
    next_node = temp.next
    temp.next = prev
    prev = temp
    temp = next_node

head = prev

temp = head
while temp:
    print(temp.data, end=" -> ")
    temp = temp.next
print("None")
```

```
30 -> 20 -> 10 -> None
```

```
# 7. Calculate the Sum of Every Two Consecutive Node Values in Python
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

head = Node(10)
head.next = Node(20)
head.next.next = Node(30)
head.next.next.next = Node(40)

temp = head

while temp and temp.next:
    print(temp.data + temp.next.data)
    temp = temp.next
```

```
30
50
70
```